MGT 155 Final Project

For the final project, you have been assigned teams based on your preferences in Canvas. An assignment has been created for you to submit your final project. Only one member of your team needs to submit it by the due date, and it will apply to your entire team. The final project will consist of two components: (i) airline overbooking, and (ii) process modeling with discrete event simulation. Your final report should be 3-5 pages long and address each of the guiding questions provided to you. This portion should be expositional whereas your technical analysis should be included separately as appendices.

Component I: Overbooking problem

This problem is similar to the inventory problem like cupcakes that we considered in class. Southwest Airlines runs daily flights from San Diego to Boston. They face a fixed cost of $70,000 for each flight independent of the actual number of passengers. There are 250 seats available on the plane. One-way tickets generate a revenue of $600 a piece when used, but are fully refundable if not used. The dataset contains how many seats are booked and the number of no shows (which is uncertain). Note that as more seats are booked, the number of no shows is also increasing on average. By law, an airline is allowed to overbook flights, but must give compensation to all ticketed passengers who are not allowed to board due to overbooking. They are auctioning the alternative transportation to the booked customers who get bumped out. The total compensation and per passenger compensation are given in the dataset (which is also uncertain). From the data, observe that if more customers are needed to be bumped out, per passenger compensation also increases on average.

In addition to providing compensation, the airline must also provide those passengers with alternative transportation on another carrier (the cost of providing the alternative transportation just wipes out the $600 revenue).

How many tickets should Southwest book on its flight from San Diego to Boston? Provide your team's answer with a clear explanation..

Component II: Process Modeling and Simulation

For the second component, you are to build a simulation model using the `simpy` library in python. Consider a bank with both cashiers and an ATM machine. The process flow is as follows. On average, 45 customers per hour enter the bank. There is a 50% chance each customer heads to the ATM first and a 50% chance the customer heads straight to the cashiers. Service at the ATM follows a triangular distribution with minimum duration of 1 minute, maximum duration of 4 minutes, and a mode duration of 2 minutes. Documentation about the triangular distribution can be found here:

`https://numpy.org/doc/2.0/reference/random/generated/numpy.random.triangular.html`

Service at the cashier also follows a triangular distribution with minimum duration of 3 minutes, maximum duration of 20 minutes, and a mode duration of 5 minutes. For customers that first go to the ATM, there is a 30% chance each customer will then go see a cashier and a 70% chance the customer will exit the bank immediately after ATM service. Customers always exit immediately after seeing a cashier. There are 5 cashiers in the bank and a single ATM machine. There is a single queue in front of the pool of cashiers. Figure 1 provides a visual illustration of process flow at the bank.

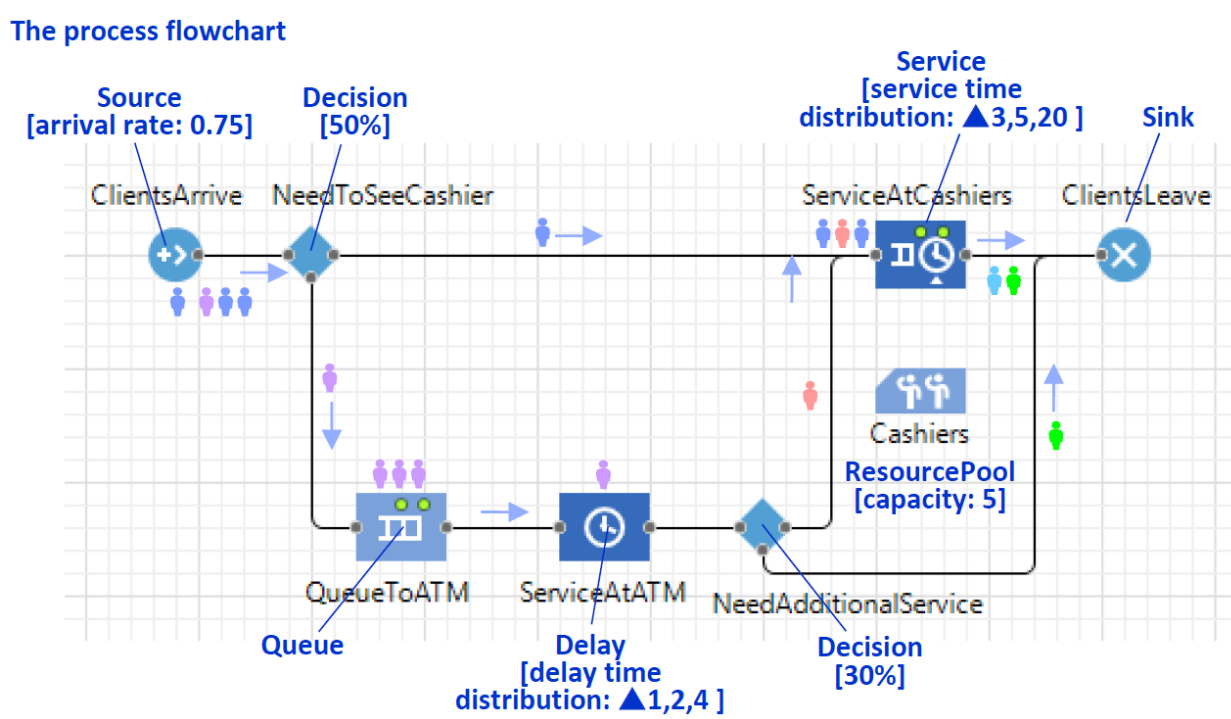

Figure 1: Process flow diagram for the bank

For this component, do the following and discuss them in your report:

- Construct a simpy model of process flow at the bank. Include the code of your model in the report appendices.
- Use your model to compute the following metrics
  - Utilization of cashiers
  - Mean queue length at cashiers
  - Mean queue length at ATM
  - Construct the distribution of time spent in the bank (Hint: the mean time in the bank should be around 17 to 18 minutes)
- Compare the following possibilities for improving operations: (i) hiring a sixth cashier, or (ii) training the 5 cashiers so that the maximum service time at a cashier becomes only 10 minutes instead of 20 minutes. Which change is more effective in your opinion? Explain why.